

Q1. JEE Main 2026 (21 January Shift 2)

By usual analysis, 1.00 g of compound (X) gave 1.79 g of magnesium pyrophosphate. The percentage of phosphorus in compound (X) is : (nearest integer) (Given, molar mass in gmol^{-1} : O = 16, Mg = 24, P = 31)

- (1) 30 (2) 50 (3) 40 (4) 20

Q2. JEE Main 2026 (28 January Shift 2)

A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is _____.
(Given Molar mass in gmol^{-1} S : 32, BaSO_4 : 233)

- (1) 48.24 % (2) 63.15 % (3) 21.05 % (4) 42.10 %

Q3. JEE Main 2026 (24 January Shift 2)

In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as

- (1) sulphide & sulphide (2) chromate & sulphide
(3) carbonate & carbonate (4) hydroxide & carbonate

Q4. JEE Main 2026 (24 January Shift 1)

Consider three metal chlorides x, y and z, where x is water soluble at room temperature, y is sparingly soluble in water at room temperature and z is soluble in hot water. x, y and z are respectively

- (1) CuCl_2 , AgCl and PbCl_2 (2) AlCl_3 , PbCl_2 and BaCl_2
(3) MgCl_2 , AgCl and AlCl_3 (4) AgCl , Hg_2Cl_2 and PbCl_2

ANSWERS AND SOLUTIONS

1. (2)

2. (3)

3. (3)

4. (1)

1. (2) Given: 1.00 g of compound (X) produces 1.79 g of $\text{Mg}_2\text{P}_2\text{O}_7$.

Molar mass of $\text{Mg}_2\text{P}_2\text{O}_7 = 2(24) + 2(31) + 7(16) = 222 \text{ g/mol}$.

Moles of $\text{Mg}_2\text{P}_2\text{O}_7 = 1.79 / 222 = 0.00806 \text{ mol}$.

Since each mole of $\text{Mg}_2\text{P}_2\text{O}_7$ contains 2 moles of P, moles of P = $2 \times 0.00806 = 0.01612 \text{ mol}$.

Mass of P = $0.01612 \times 31 = 0.4997 \text{ g} \approx 0.50 \text{ g}$.

Percentage of phosphorus = $(0.50 / 1.00) \times 100 = 50\%$.

2. (3) Given: Mass of organic compound = 0.314 g, Mass of BaSO_4 obtained = 0.4813 g.

$$\%S = \frac{32}{233} \times \frac{0.4813}{0.314} \times 100$$

$$= \frac{32 \times 0.4813 \times 100}{233 \times 0.314}$$

$$= \frac{1540.16}{73.162}$$

$$= 21.05\%$$

3. (3) In Group V of cation analysis, both Ba^{2+} and Ca^{2+} are precipitated by adding Na_2CO_3 or $(\text{NH}_4)_2\text{CO}_3$ solution.

Ba^{2+} precipitates as BaCO_3 (white precipitate) and Ca^{2+} precipitates as CaCO_3 (white precipitate). Both form insoluble carbonates under these conditions, making the answer carbonate & carbonate.

4. (1) Metal chloride solubility characteristics:

x must be water soluble at room temperature, y must be sparingly soluble at room temperature, and z must be soluble in hot water but not in cold water.

CuCl_2 is highly soluble in water (x).

AgCl is notoriously insoluble in water and sparingly soluble at any temperature, classifying it as sparingly soluble (y).

PbCl_2 exhibits unusual solubility behavior—it is sparingly soluble in cold water but its solubility increases with temperature, making it soluble in hot water (z).

This combination perfectly matches all three criteria.